MTH 765P - Mini Project Football Analytics

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1 1. Introduction

Arsenal is in the midst of a severe goal-scoring drought, managing only one goal despite 61 shots and an xG of 6.47 over the last three matches, a problem that has persisted for seven games. The urgency to secure a new striker is evident, and leveraging key statistics from the live Transfermarkt database to identify a top-class striker appears to be a fruitful initiative to address this pressing issue in the football world.

2 2. Datset

Source Transfermarkt is a prominent German based online platform focused on football, providing a vast database of players and clubs with detailed statistics and market values. It's a go-to resource for football enthusiasts to track player transfers, access match data, and stay informed about the latest football news and rumors.

The following data set , (https://www.kaggle.com/datasets/davidcariboo/player-scores/data),

is a clean, structured and automatically updated football data from Transfermarkt. The dataset is automatically updated **once a week**.

The dataset's source code and data pipeline can be found on GitHub (https://github.com/dcaribou/transfermarkt-datasets).

The project utilises a "transfermarkt-scraper" to extract data from the Transfermarkt website. Python scripts and SQL are used for data curation and publication within this repository.

As of January 2023 the dataset consists of :

- `60,000+` games
- `400+` clubs
- `30,000+` players
- `400,000+` player market valuations based on historical records

Figure 1: Structural overview

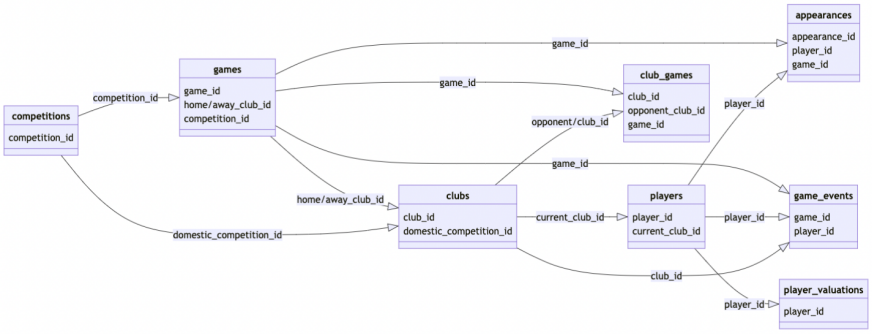


Image taken from Date Set overview on Kaggle

- `1,200,000+` player appearance statistical records from all games

Comprising several CSV files, the dataset encapsulates detailed records on competitions, games, clubs, players, and individual appearances. These files are meticulously linked with various IDs, facilitating comprehensive relational analysis

The dataset's breadth offers an invaluable resource for in-depth analysis. While its complexity may appear challenging at first glance, it is this very attribute that enables detailed and inclusive analytical opportunities.

This is a Structural overview of the dataset:-

Each dataset entry is meticulously recorded, with appearances files detailing individual player game participation, and clubs files enumerating unique club data. Keys such as playerid, gameid, and clubid are utilized for cross-referencing. The inter-connectivity of these datasets is pivotal for multifaceted analysis, enabling a holistic view of the football ecosystem.

The dataset includes the following key features and relationships:

Competitions: This section contains a unique competitionid that helps filter games, clubs, and player performances by specific competitions like the Premier League or La Liga.

Games: Individual matches are recorded with unique gameid identifiers, including details about home and away clubs (home/awayclubid) and associated competitionid. This information is crucial for analyzing matches within specific competitions.

Clubs: Clubs are represented with clubid and domesticcompetitionid, allowing for the linkage of clubs to the competitions they participate in and facilitating club-specific data analysis.

Players: Player data includes playerid and links to currentclubid, enabling player performance analysis while considering their current club affiliation.

Appearances: This section is essential for player performance analysis,

as each player appearance has a separate entry. It contains metrics and key performance indicators (e.g., goals, assists, yellowcards) linked to playerid and gameid.

Club Games: Serving as a junction table between clubs and games, this section provides context on which club played against whom (opponent/clubid) and in which games (gameid).

Game Events: Records events occurring during games with references to gameid and playerid, including goals, assists, cards, substitutions, and more.

Player Valuations: This section contains financial data linked to playerid, facilitating extensive analysis related to player transfers and valuations.

Analytical Potential and Opportunities

- **Detailed Data Queries**: The dataset uses relationships between data tables to allow for detailed searches. This means you can get specific insights on everything from how well players are doing to how much they're worth.
- **Wide-Ranging Analysis:** The setup of the dataset makes it possible to look at many different parts of football data. By connecting the right IDs from different tables, you can explore lots of angles and details.
- **Up-to-Date Information:** The data gets updated every week, which means you can always work with the latest figures. This is really important in football, where things like player performance, team changes, and player values can change quickly. With this dataset, you can keep track of these changes as they happen.

Challenges

- **Complexity of Data Relationships:** With multiple interlinked CSV files, understanding and navigating the relationships between various data points requires a solid grasp of relational database concepts. Crafting SQL queries or python code that joins these tables correctly can be intricate and time-consuming.
- **Data Volume**: The sheer size of the dataset, with hundreds of thousands of records across multiple tables, can be overwhelming. Efficient data handling and query optimization become critical to manage computational load and ensure timely retrieval of information.
- **Analytical Depth:** Deciding the level of depth for analysis can be challenging. Decision making in football can derive from a plethora of parameters.

3 3. Identifying Potential Analysis Opportunities

Some potential Analysis opportunities include:

1. Player Efficiency: Goals and assists per 90 minutes.
2. Playing Time Analysis: Correlation between minutes played and performance metrics.
3. Predictive Modelling: Estimating future goals/assists using historical data.
4. Player Ranking system: Weighted index based on multiple performance indicators.
5. Market Value Correlation: Relationship between performance stats and transfer fees.
6. Contract Analytics: Data-driven insights for player contract negotiations.

3.1 Initial Motivation

Finding the best striker for Arsenal to acquire. Using data to find a striker turns out to be more complicated than it first appears. Here are a few of the challenges that make this goal less than ideal given the comparatively limited scope for this project:

Multifaceted Assessment: Evaluating the ideal striker involves considering various factors like goals, assists, height, market value, and league competitiveness, resulting in a complex assessment process. Additionally, introducing a ranking metric as a potential solution presents challenges, as it requires careful consideration and allocation of criteria and weights for factors such as goals per match, height, age, market value, league competitiveness, and foot preference.

Financial Constraints: Budget considerations, including market value and club finances, impose limitations, impacting striker selection and available options.

League Evaluation: Assessing a player's performance in their current league introduces complexity, requiring an analysis of the level of competition faced.

Long vs. Short-Term Strategy: Choosing between long-term talent development and short-term acquisition adds another layer of complexity, which necessitates a balance between immediate impact and future sustainability.

Longevity Consideration: Factors like player age and injury proneness must also be taken into account when identifying the right striker.

3.2 Updated Motivation

Optimising data extraction on player goal scoring metrics.

Key Area of focus

My primary focus for this project is to optimize the extraction of player metrics related to goal scoring.

Upon a thorough analysis of the dataset, it's evident that there exists significant potential for in-depth analysis. However, this potential comes with complexity due to the intricate relationships within the dataset. My central

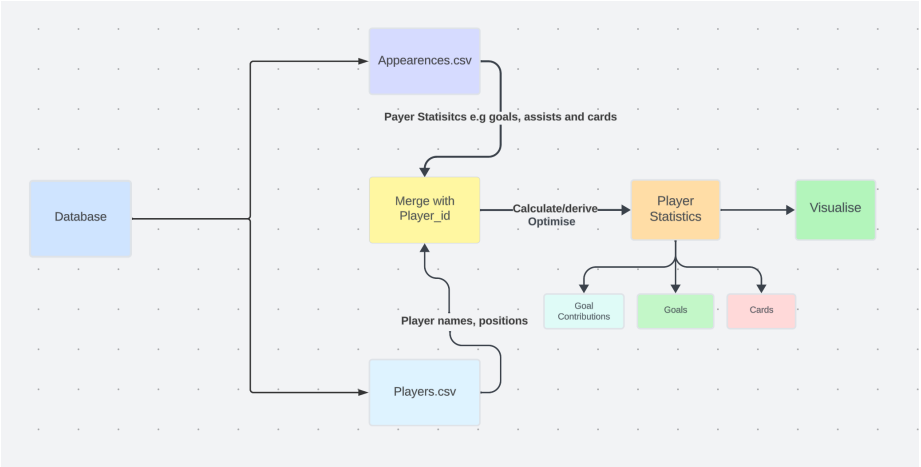


Figure 2: Process Overview

objective is to develop code that simplifies the retrieval of player goal scoring metrics, making it as convenient as possible.

Ideally, I aim to create a system where I can input a start and end date, and the code will provide me with a list of the highest goal scorers within that specific time frame.

With this streamlined approach, I intend to answer questions in real-time, utilizing the dynamic nature of the dataset. This can provide up to date responses to questions such as:

- Who scored the most goals last month?
- Who had the most goals and assists in the last decade?
- Which player contributed the most to goals between 2015-2016?

4 4. Processing the Dataset

Initially I collected football data from two main sources: 'appearances.csv' and 'players.csv.' After merging these datasets based on a common identifier, 'player_id,' we can calculate various player statistics like goals, assists, and averages.

4.1 Conducting Exploratory Data Analysis

We looked at all 9 files and found that 'appearances.csv' and 'players.csv' have the most important player statistics which are key for our objective. So, we decided to take a closer look at these two files to learn more about how players are performing.

Conducting Exploratory Data Analysis gave the following key insights for our objective:

- The large number of missing values in the "first name" column is crucial for our analysis, especially when labeling top goal scorers. This is because some players are only listed with their last name, which includes their full name. For instance, "Mohammed Salah" has his full name in the "last_name" column. Therefore, we will use the "last_name" column for our visualization to ensure accurate results.
- We examined the time span covered by our dataset, which at the moment ranges from July 3, 2012, to January 11, 2024,.This time range informs us about the constraints in using the data to uncover statistics and insights, as we cannot analyze events that occurred before this period.

After merging these datasets based on a common identifier, 'player_id,' we can calculate various player statistics like goals, assists, and averages.

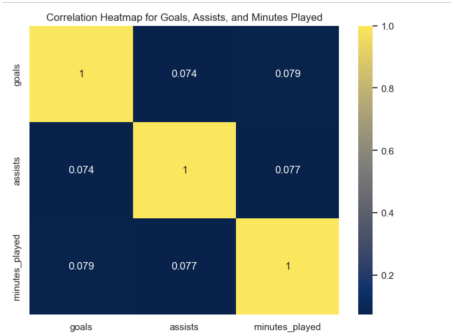
5 Initial Insights on player statistics

1. Correlation Heat-map for Goals, Assists, and Minutes Played

Overall, the correlations between these variables are positive but very low which might seem as a surprise. At first glance you might think players who play more will score or assist more and there might be a strong correlation between goals and assists but that is not the case. This reflects the complex nature of the sport.

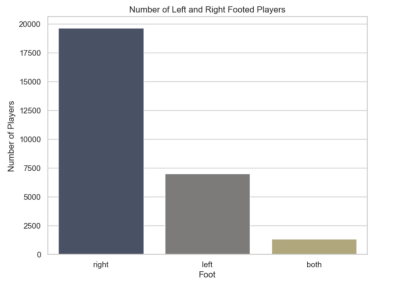
There is no strong linear relationship between the number of goals and assists a player makes, and the minutes they play. This could mean that while playing time is necessary to have opportunities to score or assist, it is not the only factor that will determine a player's effectiveness in these areas.

Of course there are so many other variables that might also play significant roles such as position, age, height to name a few.



2. Distribution of dominant playing foot As expected Right footed

players are significantly greater than left footed ones. Two footed players are extremely rare.



6 6. Code generation for optimising extraction of Statistical Figures

6.1 Version 1

We begin by setting a time threshold six months prior to the current date and then filter player appearance records to only include those within this time frame. The 'date' column is converted to datetime format to facilitate this filtering.

Once the data is filtered, we aggregates the total goals scored by each player, merge these figures with a player dataset to associate player names with their respective goal tallies.

The data is then sorted in descending order of goals scored, and we finally display the top 10 scorers.

6.1.1 Shortcomings

We can input any time period to subtract from todays date but the limitation of this code lies in its inflexibility to adjust to different time ranges. It relies on the current date.

6.2 Version 2

We define a function, top_scorers_in_timeframe, which identifies the top football goal scorers within a user-specified timeframe.

It begins by converting user-inputted start and end dates from strings to datetime objects.

The function then filters player appearance records to include only those that fall between these dates.

Following the filtering process, we aggregate the total number of goals scored by each player during this period

Then we merge this data with a separate player dataset to match player IDs with their names.

Then we sort the merged data to highlight the players with the most goals, and the top 10 scorers are returned.

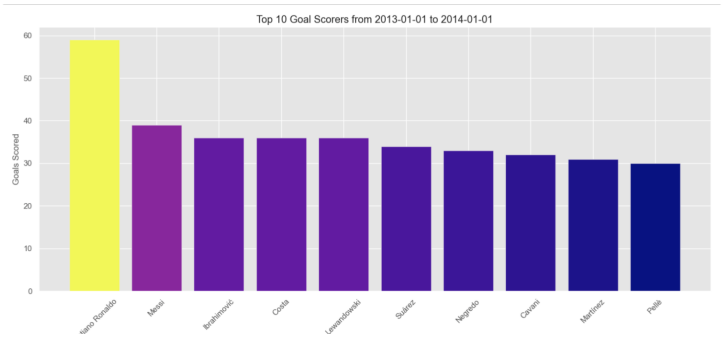
The user-defined input capability significantly enhances the function's applicability and optimises the data extraction process.

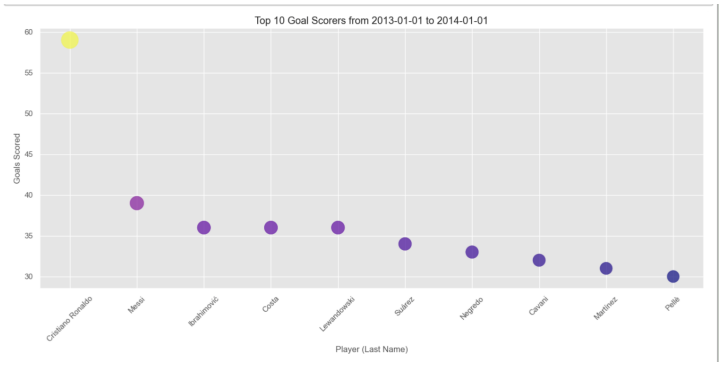
```
Enter the start date (YYYY-MM-DD): 2013-01-01
Enter the end date (YYYY-MM-DD): 2014-01-01
```

7 Visualisation

7.1 Goals Analysis

Following optimising the data extraction process we can now visualise our goal scoring metrics. We can change inputs to get highest goalscorers in the last decade, year or week. For the input of dates above we get the following visualisations:





8 Future Potential and Scope

Create a web interface, possibly utilizing Streamlit, to execute this code. This interface will offer convenient access to optimized football statistics, including top goal scorers and assist leaders, for any specified time frame. Given the dynamic nature of the dataset, it enables users to generate statistics within seconds.